PVT CORNER TIMING ANALYSIS

VSDBabySoC Design | Static Timing Analysis with OpenSTA

1. Aim of the Experiment

This study evaluates the timing behavior of the VSDBabySoC design across different Process, Voltage, and Temperature (PVT) operating conditions. OpenSTA reports are used to compare timing margins and identify corners where timing requirements are not met.

2. Background

Static Timing Analysis (STA) checks signal-path delays against the design's timing requirements without simulating every possible input sequence. In this report, sixteen PVT corners are compared using minimum slack, maximum slack, Worst Negative Slack (WNS), Total Negative Slack (TNS), and the reported status.

3. PVT Corner Results

No.	PVT corner	Min slack (ns)	Max slack (ns)	WNS (ns)	TNS (ns)	Status
1	FF_100C_1v65	0.249	4.055	0.000	0.000	MET
2	FF_100C_1v95	0.196	5.567	0.000	0.000	MET
3	FF_n40C_1v56	0.292	2.278	0.000	0.000	MET
4	FF_n40C_1v65	0.255	3.411	0.000	0.000	MET
5	FF_n40C_1v76	0.224	4.430	0.000	0.000	MET
6	FF_n40C_1v95	0.187	5.592	0.000	0.000	MET
7	SS_100C_1v40	0.905	-17.122	-17.122	-2599.514	VIOLATED
8	SS_100C_1v60	0.642	-7.874	-7.874	-892.705	VIOLATED
9	SS_n40C_1v28	1.830	-62.563	-62.563	-13957.588	VIOLATED
10	SS_n40C_1v35	1.348	-39.697	-39.697	-7993.793	VIOLATED
11	SS_n40C_1v40	1.125	-29.694	-29.694	-5466.484	VIOLATED
12	SS_n40C_1v44	0.991	-23.908	-23.908	-4107.187	VIOLATED
13	SS_n40C_1v60	0.663	-10.609	-10.609	-1355.797	VIOLATED
14	SS_n40C_1v76	0.504	-4.382	-4.382	-367.928	VIOLATED
15	TT_025C_1v80	0.310	1.939	0.000	0.000	MET
16	TT_100C_1v80	0.314	2.094	0.000	0.000	MET

The detailed timing graphs and path reports follow on the subsequent pages. The table reproduces the reported values from the source report.

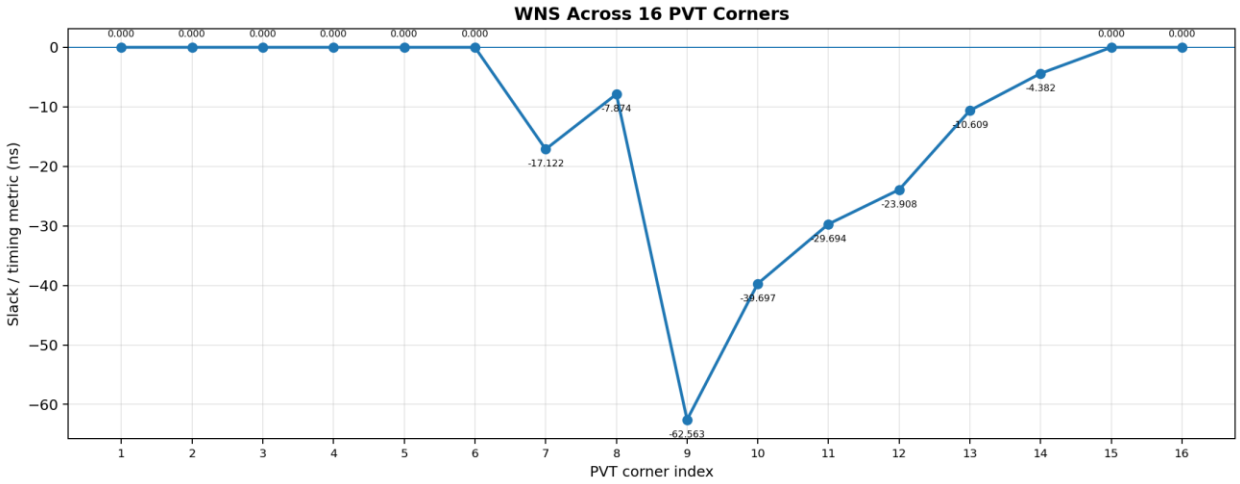


Figure 1: Worst Negative Slack (WNS) across PVT corners

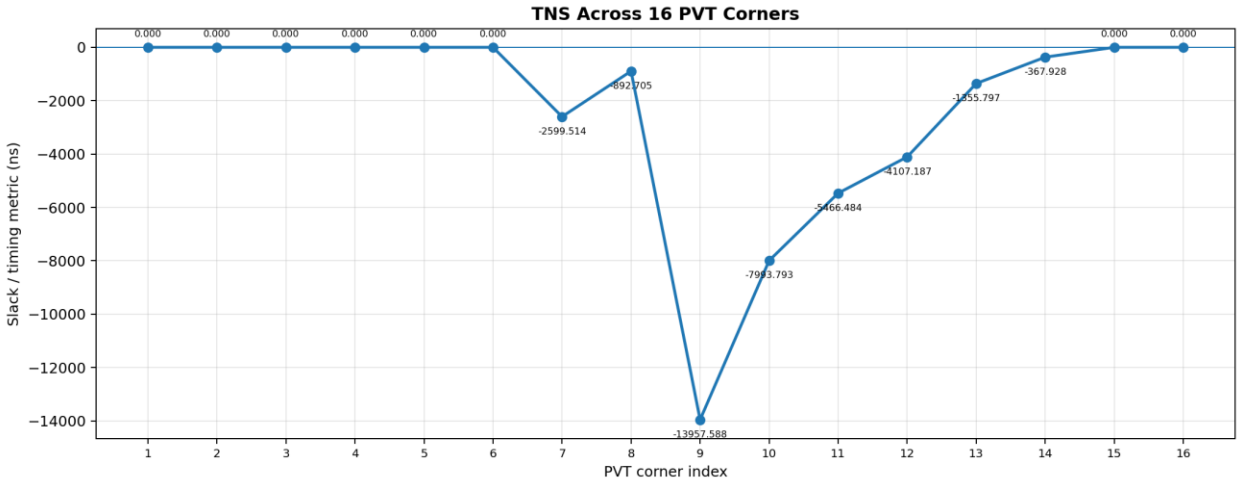


Figure 2: Total Negative Slack (TNS) across PVT corners

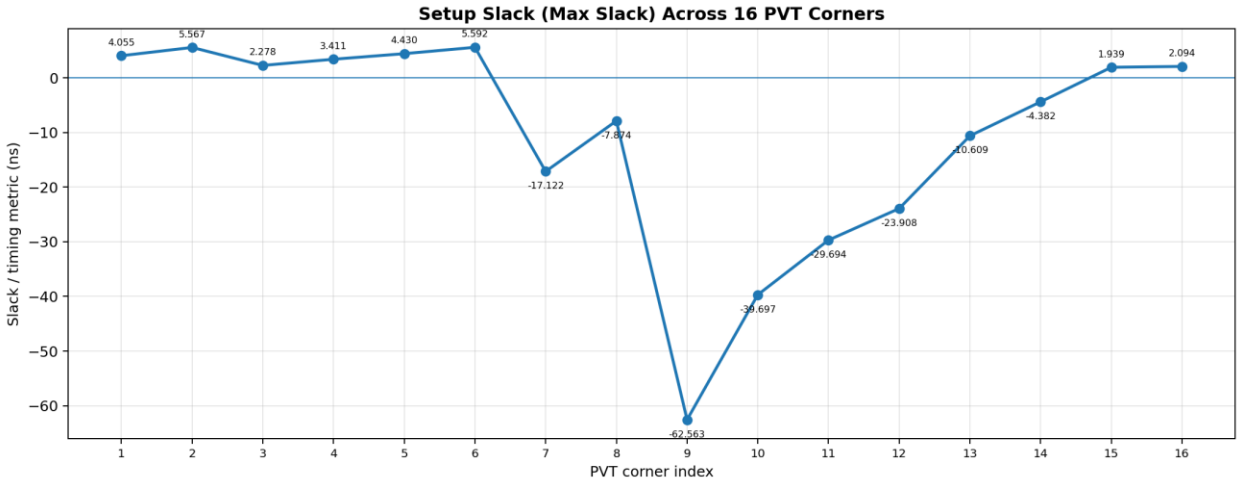


Figure 3: Setup Slack across PVT corners

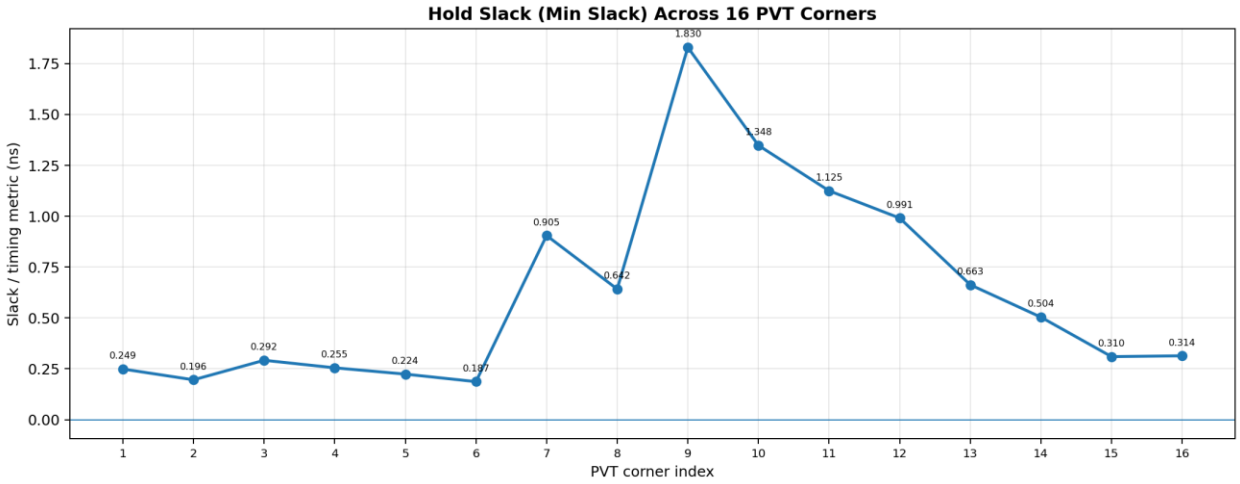


Figure 4: Hold Slack across PVT corners

5. STA REPORTS

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.24 0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.09 0.34 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.08 0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.42 0.83 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.13 0.96 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.19 1.16 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.25 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.15 1.40 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.49 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.15 1.64 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.73 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.15 1.89 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.98 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.13 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.22 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.46 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.61 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.71 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.86 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.95 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.14 3.09 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.28 3.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.14 3.50 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.26 3.76 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.27 4.03 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.27 4.31 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.28 4.59 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.04 4.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.05 4.68 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.26 4.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.27 5.21 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.27 5.48 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.29 5.77 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.14 5.91 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.26 6.17 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.38 6.55 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.20 6.75 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.13 6.88 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
0.00 6.88 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
6.88 data arrival time

10.93 data required time

-6.88 data arrival time

4.05 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_100C_1v95

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.20 0.20 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.07 0.27 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.07 0.33 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.35 0.68 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.09 0.77 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.15 0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.07 0.99 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.18 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.67 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.86 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.12 2.05 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.12 2.24 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.31 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.11 2.42 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.22 2.63 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.11 2.74 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.20 2.95 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.16 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.37 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.22 3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.03 3.61 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 3.66 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.22 4.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.11 4.62 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.20 4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.30 5.12 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.17 5.28 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.11 5.39 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
0.00 5.39 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
5.39 data arrival time

10.96 data required time

-5.39 data arrival time

5.57 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v56

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.27 0.27 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.13 0.39 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.08 0.48 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.54 1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.15 1.16 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.27 1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.10 1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.21 1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.10 1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.10 2.14 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.35 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.10 2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.65 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.10 2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.96 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.10 3.06 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.26 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.10 3.36 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.57 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.10 3.67 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.17 3.84 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.35 4.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.18 4.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.33 4.71 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.35 5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.35 5.41 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.34 5.74 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.06 5.80 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.05 5.85 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.34 6.18 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.35 6.53 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.35 6.88 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.36 7.24 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.18 7.42 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.33 7.76 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.34 8.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.22 8.32 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.16 8.48 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.16 8.64 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
0.00 8.64 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
8.64 data arrival time

10.92 data required time

-8.64 data arrival time

2.28 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v65

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.24 0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.11 0.35 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.07 0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.48 0.90 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.12 1.02 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.23 1.26 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.34 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.18 1.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.61 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.18 1.79 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.88 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.15 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.33 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.41 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.59 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.68 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.86 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.95 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.18 3.13 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.09 3.22 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.15 3.37 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.30 3.67 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.16 3.83 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.29 4.12 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.30 4.42 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.30 4.72 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.29 5.01 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.05 5.06 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 5.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.29 5.39 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.30 5.69 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.30 5.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.31 6.30 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.16 6.46 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.29 6.75 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.30 7.05 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.19 7.24 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.14 7.38 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.14 7.52 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
0.00 7.52 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
7.52 data arrival time

10.93 data required time

-7.52 data arrival time

3.41 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[1][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v76

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.21 0.21 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.10 0.31 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.06 0.37 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.43 0.81 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.09 0.90 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.20 1.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.17 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.16 1.33 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.41 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.16 1.56 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.64 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.16 1.80 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.87 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.03 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.26 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.33 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.49 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.56 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.72 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.07 2.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.13 2.93 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.26 3.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.14 3.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.25 3.58 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.26 3.83 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.26 4.09 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.25 4.34 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.04 4.38 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 4.42 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.25 4.68 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.26 4.93 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.26 5.19 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.27 5.46 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.14 5.60 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.25 5.85 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.25 6.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.17 6.27 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.12 6.39 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.12 6.52 ^ core/U643/Y (sky130_fd_sc_hd__o22ai_1)
0.00 6.52 ^ core/CPU_Xreg_value_a4_reg[1][31]/D (sky130_fd_sc_hd__dfxtp_1)
6.52 data arrival time

10.95 data required time

-6.52 data arrival time

4.43 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v95

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.18 0.18 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.08 0.26 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.05 0.32 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.37 0.69 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.06 0.75 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.17 0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.06 0.98 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.17 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.68 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.87 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.13 2.06 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.06 2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.13 2.26 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.06 2.32 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.11 2.43 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.21 2.64 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.11 2.75 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.20 2.96 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.17 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.38 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.20 3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.03 3.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 3.65 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.22 4.50 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.11 4.61 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.20 4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.21 5.02 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.14 5.17 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.10 5.27 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.10 5.37 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
0.00 5.37 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
5.37 data arrival time

10.96 data required time

-5.37 data arrival time

5.59 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_100C_1v40

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.91 0.91 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.21 1.13 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
0.18 1.30 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
0.71 2.01 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
0.80 2.81 v core/U478/X (sky130_fd_sc_hd__a22o_1)
0.47 3.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.58 3.86 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.36 4.22 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.48 4.70 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.36 5.05 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.48 5.53 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.36 5.89 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.48 6.37 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.36 6.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)

0.48 7.20 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.36 7.56 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.48 8.04 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.36 8.39 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.48 8.87 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.36 9.23 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.48 9.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.36 10.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.49 10.55 v core/U96/X (sky130_fd_sc_hd__a21o_1)
1.32 11.87 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.51 12.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)
1.27 13.65 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
1.33 14.98 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
1.34 16.32 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
1.13 17.45 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.13 17.58 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.17 17.75 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1.30 19.05 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
1.33 20.38 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
1.33 21.70 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
1.37 23.08 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.51 23.59 v core/U81/X (sky130_fd_sc_hd__a21o_1)
1.27 24.86 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
1.63 26.49 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.66 27.15 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.39 27.54 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 27.54 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
27.54 data arrival time

10.42 data required time

-27.54 data arrival time

-17.12 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_100C_1v60

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.62 0.62 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.15 0.78 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
0.13 0.91 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
0.51 1.42 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
0.56 1.98 v core/U478/X (sky130_fd_sc_hd__a22o_1)
0.33 2.31 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.42 2.72 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.24 2.97 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.34 3.31 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.24 3.55 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.34 3.89 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)

0.24 4.13 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.34 4.47 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.24 4.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.34 5.06 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.30 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.34 5.64 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.88 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.34 6.22 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.24 6.46 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.34 6.81 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.24 7.05 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.34 7.38 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.86 8.24 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.35 8.59 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.83 9.42 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.86 10.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.87 11.14 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.74 11.89 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.09 11.98 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.12 12.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.84 12.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.86 13.80 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.86 14.66 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.89 15.55 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.35 15.90 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.83 16.73 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
1.04 17.77 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.48 18.25 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.26 18.51 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 18.51 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
18.51 data arrival time

10.64 data required time

-18.51 data arrival time

-7.87 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v28

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
1.50 1.50 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.71 2.20 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.56 2.76 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
2.67 5.43 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
1.71 7.14 v core/U479/X (sky130_fd_sc_hd__xor2_1)
1.42 8.57 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.76 9.33 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
1.20 10.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)

0.76 11.29 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
1.20 12.49 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.76 13.25 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
1.20 14.45 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.76 15.21 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
1.20 16.41 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.76 17.17 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
1.20 18.38 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.76 19.14 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
1.20 20.34 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.76 21.10 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
1.20 22.30 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.76 23.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
1.13 24.19 v core/U96/X (sky130_fd_sc_hd__a21o_1)
3.70 27.89 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
1.32 29.21 v core/U94/X (sky130_fd_sc_hd__a21o_1)
3.62 32.83 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
3.78 36.61 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
3.80 40.42 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
3.39 43.81 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.40 44.21 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.21 44.43 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
3.63 48.06 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
3.78 51.84 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
3.78 55.62 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
3.86 59.48 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
1.32 60.80 v core/U81/X (sky130_fd_sc_hd__a21o_1)
3.62 64.42 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
4.76 69.19 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
1.61 70.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.91 71.70 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 71.70 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
71.70 data arrival time

9.14 data required time

-71.70 data arrival time

-62.56 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v35

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
1.12 1.12 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.52 1.64 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.41 2.05 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
2.01 4.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
1.22 5.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
1.04 6.33 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)

0.54 6.87 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.87 7.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.54 8.28 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.87 9.15 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.54 9.69 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.87 10.56 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.54 11.10 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.87 11.96 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.54 12.51 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.87 13.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.54 13.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.87 14.78 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.54 15.33 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.87 16.19 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.54 16.74 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.81 17.54 v core/U96/X (sky130_fd_sc_hd__a21o_1)
2.47 20.01 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.91 20.92 v core/U94/X (sky130_fd_sc_hd__a21o_1)
2.42 23.34 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
2.52 25.87 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
2.54 28.40 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
2.27 30.67 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.27 30.94 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.17 31.11 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
2.43 33.54 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
2.52 36.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
2.52 38.59 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
2.57 41.16 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.91 42.07 v core/U81/X (sky130_fd_sc_hd__a21o_1)
2.42 44.49 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
3.14 47.63 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
1.20 48.83 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.64 49.47 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 49.47 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
49.47 data arrival time

9.77 data required time

-49.47 data arrival time

-39.70 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v40

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.94 0.94 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.43 1.38 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.34 1.72 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
1.70 3.42 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)

0.99 4.40 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.87 5.27 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.44 5.71 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.71 6.43 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.44 6.87 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.71 7.58 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.44 8.02 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.71 8.74 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.44 9.18 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.71 9.89 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.44 10.33 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.71 11.05 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.44 11.49 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.71 12.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.44 12.64 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.71 13.36 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.44 13.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.66 14.46 v core/U96/X (sky130_fd_sc_hd__a21o_1)
1.95 16.41 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.74 17.14 v core/U94/X (sky130_fd_sc_hd__a21o_1)
1.91 19.05 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
1.98 21.04 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
2.00 23.03 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
1.79 24.82 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.22 25.04 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.14 25.18 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1.92 27.10 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
1.98 29.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
1.98 31.07 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
2.02 33.10 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.74 33.83 v core/U81/X (sky130_fd_sc_hd__a21o_1)
1.91 35.74 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
2.46 38.20 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
1.00 39.20 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.53 39.74 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 39.74 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
39.74 data arrival time

10.04 data required time

-39.74 data arrival time

-29.69 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v44

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.83 0.83 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.38 1.21 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)

0.30 1.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
1.50 3.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.88 3.88 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.77 4.65 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.39 5.04 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.62 5.67 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.39 6.06 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.62 6.68 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.39 7.07 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.62 7.69 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.39 8.08 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.62 8.71 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.39 9.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.62 9.72 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.39 10.11 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.62 10.73 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.39 11.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.62 11.75 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.39 12.14 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.57 12.71 v core/U96/X (sky130_fd_sc_hd__a21o_1)
1.65 14.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.63 15.00 v core/U94/X (sky130_fd_sc_hd__a21o_1)
1.61 16.61 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
1.67 18.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
1.68 19.96 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
1.51 21.48 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.19 21.67 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.13 21.79 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1.62 23.42 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
1.67 25.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
1.67 26.76 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
1.71 28.47 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.63 29.10 v core/U81/X (sky130_fd_sc_hd__a21o_1)
1.61 30.72 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
2.07 32.79 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.87 33.65 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.45 34.10 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 34.10 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
34.10 data arrival time

10.20 data required time

-34.10 data arrival time

-23.91 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v60

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)

0.57 0.57 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.25 0.82 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.20 1.02 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
1.04 2.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.53 2.58 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.52 3.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.24 3.35 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.41 3.76 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.24 4.00 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.41 4.41 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.24 4.66 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.41 5.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.31 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.41 5.72 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.96 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.41 6.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.24 6.62 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.41 7.03 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.24 7.27 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.41 7.68 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.25 7.93 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.37 8.30 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.98 9.28 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.40 9.68 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.96 10.64 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.99 11.63 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
1.00 12.63 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.91 13.53 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.12 13.65 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.09 13.74 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.96 14.71 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.99 15.70 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.99 16.69 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
1.02 17.71 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.40 18.11 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.96 19.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
1.21 20.27 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.59 20.87 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.29 21.15 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 21.15 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
21.15 data arrival time

10.54 data required time

-21.15 data arrival time

-10.61 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v76

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.43 0.43 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.19 0.62 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.15 0.77 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.80 1.57 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.36 1.93 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.39 2.32 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.18 2.50 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.31 2.80 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.18 2.98 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.31 3.29 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.18 3.47 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.31 3.77 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.18 3.95 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.31 4.26 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.18 4.44 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.31 4.74 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.18 4.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.31 5.23 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.18 5.41 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.31 5.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.18 5.89 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.27 6.16 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.68 6.84 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.29 7.13 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.65 7.79 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.68 8.47 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.68 9.15 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.63 9.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.09 9.87 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.07 9.94 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.66 10.61 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.68 11.28 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.68 11.96 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.70 12.66 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.29 12.95 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.65 13.61 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.82 14.43 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.45 14.88 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.19 15.07 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 15.07 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
15.07 data arrival time

10.69 data required time

-15.07 data arrival time

-4.38 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: TT_025C_1v80

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.28 0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.12 0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.10 0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.51 1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.18 1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.25 1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.11 1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.19 1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.11 1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.19 3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.19 3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.17 3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.39 4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.18 4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.37 4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.38 5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.38 5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.36 5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.05 5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.06 5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.37 6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.38 6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.38 7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.40 7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.18 7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.37 8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.45 8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.29 8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.10 8.90 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 8.90 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
8.90 data arrival time

10.84 data required time

-8.90 data arrival time

1.94 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk

Path Type: max
Corner: TT_100C_1v80

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.29 0.29 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.11 0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.10 0.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.48 0.99 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.19 1.18 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.23 1.41 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.12 1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.18 1.71 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.11 1.83 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.01 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.12 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.31 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.42 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.61 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.72 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.90 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.02 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.18 3.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.32 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.18 3.50 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.62 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.17 3.79 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.38 4.16 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.17 4.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.36 4.69 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.37 5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.37 5.43 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.35 5.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.05 5.83 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.06 5.89 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.36 6.26 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.37 6.62 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.37 6.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.39 7.38 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.17 7.55 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.36 7.91 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.49 8.40 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.25 8.65 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.16 8.81 ^ core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 8.81 ^ core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
8.81 data arrival time

10.90 data required time

-8.81 data arrival time

2.09 slack (MET)

6. Key Findings

1. Each of the six fast-process (FF) corners is marked as meeting the reported timing checks.

2. Both typical-process (TT) cases also meet the checks shown in the report.

3. Negative WNS and TNS values appear in all eight slow-process (SS) corners, indicating setup timing issues.

4. The most severe reported result is at **SS_n40C_1v28**, where WNS reaches **-62.563 ns**.

5. The comparison suggests that delay becomes more challenging in the slow-process cases, particularly at lower supply voltages.

6. Timing margins vary with the combination of process, voltage, and temperature, so a single corner cannot represent the full operating range.

7. Conclusion

OpenSTA analysis was carried out for the VSDBabySoC design at sixteen PVT corners. The FF and TT cases meet the timing checks listed in the report, whereas every SS case shows setup timing violations. Further work should focus on the critical paths, logic delay, cell selection, and appropriate buffering. The complete multi-corner analysis should then be repeated to determine whether timing performance has improved.